Strong Holevo additivity for qubit channels

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1 Problem and notation

All Hilbert spaces are finite-dimensional complex spaces. Write $M_d = M_d(\mathbb{C})$ and let X^* denote the Hermitian adjoint. A channel is a complex-linear completely positive, trace-preserving map. For $X \succeq 0$, set

$$\Psi(X) = \operatorname{Tr}(X \log X), \qquad 0 \log 0 = 0. \tag{1.1}$$

The entropy of a density matrix is $S(\rho) = -\Psi(\rho)$; all logarithms are natural. For a unit vector ψ, write $\pi_\psi = |\psi\rangle\langle\psi|$. Define the output-entropy convex roof and optimized Holevo information by

$$\widehat{S}_\Phi(\rho) = \inf_{\rho=\sum_i p_i \pi_{\psi_i}} \sum_i p_i S(\Phi(\pi_{\psi_i})), \tag{1.2}$$

$$\chi_\Phi(\rho) = S(\Phi(\rho)) - \widehat{S}_\Phi(\rho), \qquad \chi(\Phi) = \sup_\rho \chi_\Phi(\rho). \tag{1.3}$$

The infimum ranges over all finite pure-state ensembles, with nonnegative weights summing to one. Spectral decomposition makes this set nonempty, and the entropy bounds make its infimum finite. Refining mixed signals into pure signals and using entropy concavity identifies (1.3) with the usual ensemble definition of Holevo information. No attainment of an infimum or supremum is assumed.

Theorem 1.1 (Strong Holevo additivity)**.** *Let $\Phi : M_2 \to M_2$ and $\Omega : M_n \to M_m$ be channels, where n, m are arbitrary positive integers. For every density matrix ω on $\mathbb{C}^2 \otimes \mathbb{C}^n$, with marginals ω_A, ω_B,*

$$\widehat{S}_{\Phi\otimes\Omega}(\omega) \geq \widehat{S}_\Phi(\omega_A) + \widehat{S}_\Omega(\omega_B). \tag{1.4}$$

Consequently,

$$\chi(\Phi \otimes \Omega) = \chi(\Phi) + \chi(\Omega). \tag{1.5}$$

No unitality, pure-output, Kraus-rank, real-coefficient, or symmetry condition is imposed on Φ.

This manuscript presents a proposed proof of Theorem 1.1. Sections 2–7 establish a four-block matrix entropy inequality. Sections 8–11 connect it to arbitrary qubit-to-qubit channels and finite-dimensional partners. Singular matrices and zero weights are included throughout.

The argument uses finite-dimensional spectral calculus, operator monotonicity of powers in $[0, 1]$ and of the logarithm, the implicit function theorem, and standard entropy properties. Finite Kraus and Stinespring representations follow from positivity of the Choi matrix: factor that matrix into rank-one terms, reshape the vectors into Kraus operators, and attach an orthonormal environment label to those operators. Trace preservation gives the isometry condition.

2 Positive matrix curves

For positive definite matrices, the weighted geometric mean is

$$A\#_t B = A^{1/2}(A^{-1/2}BA^{-1/2})^t A^{1/2}. \tag{2.1}$$

We use its congruence invariance and joint order monotonicity. For the midpoint mean, monotonicity follows from operator monotonicity of the square root in the second argument and the symmetry $A\#B = B\#A$; both facts also follow from its defining Riccati identity. These are standard finite-dimensional matrix facts.

Lemma 2.1 (Positive matrix curves)**.** *Let $F : [0,1] \to \mathrm{Herm}_{++}(d)$ be a smooth curve. If $F'' \succeq F'F^{-1}F'$, then $F(t) \preceq F(0)\#_t F(1)$ for every $t \in [0,1]$.*

Proof. For $\varepsilon > 0$, let $E(t) = e^{-\varepsilon t(1-t)}F(t)$. Multiplying a curve by e^{φ} gives the exact identity

$$E'' - E'E^{-1}E' = e^{\varphi}(F'' - F'F^{-1}F' + \varphi'' F) \succeq 2\varepsilon E. \tag{2.2}$$

We need a uniform local expansion. Normalize at a fixed t by congruence so that $E(t) = I$, and write $B = E'(t)$, $C = E''(t)$ in these coordinates. Then

$$\begin{aligned} E(t \pm h) &= I \pm hB + \tfrac{1}{2}h^2 C + O(h^3), \\ E(t-h)\#E(t+h) &= I + \tfrac{1}{2}h^2(C - B^2) + O(h^3). \end{aligned} \tag{2.3}$$

The second expansion follows by substituting into $XE(t-h)^{-1}X = E(t+h)$, which uniquely defines the positive midpoint mean. Undoing the congruence yields

$$E(t-h)\#E(t+h) = E(t) + \tfrac{1}{2}h^2(E'' - E'E^{-1}E')(t) + O(h^3). \tag{2.4}$$

Positivity and smoothness on the compact interval make the error uniform for $h \le t \le 1-h$. Thus, for sufficiently small h,

$$E(t) \prec E(t-h)\#E(t+h). \tag{2.5}$$

Let $G(t) = F(0)\#_t F(1)$, and take a uniform grid $t_i = i/N$ fine enough for the strict inequality. The geometric-mean curve satisfies $G(t_{i-1})\#G(t_{i+1}) = G(t_i)$: normalize its endpoints to I, B and use commuting powers of B.

Let α be the maximum, over the grid, of the least positive number with $E(t_i) \preceq \alpha G(t_i)$. The endpoints give value one. If $\alpha > 1$, it is achieved at an interior index. Monotonicity gives

$$E(t_i) \prec E(t_{i-1})\#E(t_{i+1}) \preceq \alpha\big(G(t_{i-1})\#G(t_{i+1})\big) = \alpha G(t_i). \tag{2.6}$$

This contradicts minimality of α at that index. Therefore $E \preceq G$ on every sufficiently fine grid. Taking grids with $N \to \infty$, continuity gives the inequality on the interval. Finally let $\varepsilon \downarrow 0$. No commutation of the endpoints is required. □

3 The stationary Gram identity

Split $\mathbb{C}^d = \mathbb{C}^r \oplus \mathbb{C}^s$. On the complex Hilbert-Schmidt space $\mathcal{H} = M_d(\mathbb{C})$, use $\langle H, K\rangle = \operatorname{Tr} H^*K$. Let $\mathcal{B} = M_r \oplus M_s$, $V : \mathcal{B} \hookrightarrow \mathcal{H}$ the isometric inclusion, and $V^* = \Gamma$ the diagonal-block compression, with codomain $\mathcal{B}$. Define

$$L_X(H) = XH, \quad R_X(H) = HX, \quad Q_X(t) = L_X^t R_X^{1-t}, \quad \Omega_X = \int_0^1 Q_X(t)\,dt. \tag{3.1}$$

Assume $T \succ W \succ 0$, $W \in \mathcal{B}$, and

$$\log W = \Gamma \log T + \Lambda, \qquad \Lambda = \lambda_0 I_r \oplus \lambda_1 I_s, \qquad P = \Gamma T - W. \tag{3.2}$$

Here P, W and their multiplication operators preserve $\mathcal{B}$. Let $C = Q_T(t)$, $B = Q_W(t)|_{\mathcal{B}}$, and

$$\begin{gathered} K = L_{\log T} - R_{\log T}, \qquad k = (L_{\log W} - R_{\log W})|_{\mathcal{B}}, \\ F = V^*CV - B. \end{gathered} \tag{3.3}$$

Centrality of Λ gives $V^*KV = k$. Also $C' = CK$, $B' = Bk$, and K, C, as well as k, B, commute. All four operators are self-adjoint.

Operator monotonicity of powers in $[0, 1]$ gives $Q_T(t) \succ Q_W(t)$. One can check the order by the tensor identity

$$A \otimes B - C \otimes D = (A - C) \otimes B + C \otimes (B - D) \succeq 0 \tag{3.4}$$

for ordered positive factors; strictness follows from $T \succ W$, including the endpoints. Since Q_W preserves both diagonal and off-diagonal block spaces,

$$D = C - VBV^* \succ 0, \qquad F \succ 0. \tag{3.5}$$

The following is an identity on $\mathcal{B} \oplus \mathcal{B}$, not a replacement of compressed powers by powers of a compression:

$$\begin{pmatrix} F & F' \\ F' & F'' \end{pmatrix} = [V \;\; KV]^* D [V \;\; KV] \succeq 0. \tag{3.6}$$

Indeed its blocks are, in order, $V^*CV - B$, $V^*CKV - Bk$, their common adjoint, and $V^*CK^2V - kBk$. They equal F, F', F', F'', because $kBk = Bk^2$. The Schur complement and Lemma 2.1 now yield

$$F(t) \preceq F(0) \#_t F(1) = R_P^{1-t} L_P^t|_{\mathcal{B}} = Q_P(t)|_{\mathcal{B}}. \tag{3.7}$$

The equality uses $F(0) = R_P|_{\mathcal{B}}$, $F(1) = L_P|_{\mathcal{B}}$, and commutation of left and right multiplication. Therefore

$$V^* Q_T(t) V \preceq Q_P(t)|_{\mathcal{B}} + Q_W(t)|_{\mathcal{B}} \tag{PW}$$

and integration gives $V^*\Omega_T V \preceq \Omega_P|_{\mathcal{B}} + \Omega_W|_{\mathcal{B}}$ (KM). The power comparisons are inequalities on the complex Hilbert–Schmidt space. Individual $Q_X(t)$ need not preserve Hermitian matrices. Their integral Ω_X does, so the integrated inequality restricts to the real Hermitian subspace used for Hessians below.

4 The outer minimization

Let $\Psi(X) = \operatorname{Tr} X \log X$, extended continuously to positive semidefinite matrices. Fix $\lambda_0, \lambda_1 < 0$. For $J \succeq 0$, define

$$\begin{aligned} f_J(W) &= \Psi(W) - \Psi(J+W) - \operatorname{Tr} \Lambda W, \\ G_\Lambda(J) &= \Psi(\Gamma J) + \min_{W \in \mathcal{B},\ W \succeq 0} f_J(W). \end{aligned} \tag{4.1}$$

Coercivity. Put $a = \min\{-\lambda_0, -\lambda_1\} > 0$, $b = \operatorname{Tr} J$, and $R = \operatorname{Tr} W$. Integration of the derivative of $\Psi(W + tJ)$, first with a positive regularization and then by continuity, gives

$$\Psi(J+W) - \Psi(W) \le b\{1 + \log(R + b + 1)\}. \tag{4.2}$$

Indeed $W + tJ \preceq (R + b + 1)I$, and the logarithm is operator monotone. Thus

$$f_J(W) \ge aR - b\{1 + \log(R + b + 1)\} \longrightarrow +\infty. \tag{4.3}$$

The feasible set is closed and the objective is continuous. A minimum exists. The same bound is uniform when J ranges over a compact set; $f_J(0) = -\Psi(J)$ is locally bounded. Consequently all relevant minimizers lie in one compact set. The minimum value is continuous in J, by the usual compact minimum argument.

Derivative and convexity. On positive definite matrices,

$$D \log X[H] = \Omega_X^{-1}(H) = \int_0^\infty (X + uI)^{-1} H (X + uI)^{-1}\, du. \tag{4.4}$$

This identity follows in an eigenbasis: the inverse of the logarithmic mean is $(\log a - \log b)/(a - b)$, with diagonal value $1/a$. In particular $D^2\Psi(X) = \Omega_X^{-1}$. On $\mathcal{B}$,

$$D_W^2 f_J = \Omega_W^{-1}|_{\mathcal{B}} - V^* \Omega_{J+W}^{-1} V. \tag{4.5}$$

Since $J + W \succeq W$, the tensor order used above and integration show $\Omega_{J+W} \succeq \Omega_W$. Inversion reverses order. The Hessian is therefore nonnegative. The objective is convex on the positive definite cone and, by continuity, on its closure. If $J \succ 0$, the order is strict, so the Hessian is positive definite.

Interiority for $J \succ 0$. If a minimizing W has a kernel, take its nonzero kernel projection $E \in \mathcal{B}$. Replacing W by $W + \varepsilon E$ adds $\varepsilon \log \varepsilon \operatorname{Tr} E$ to $\Psi(W)$. The other terms change by $O(\varepsilon)$, since $J + W \succ 0$. The objective decreases for small ε, a contradiction.

Thus the minimizer is positive definite and is unique by strict convexity. Its stationary equation is

$$\log W - \Gamma \log(J + W) = \Lambda. \tag{4.6}$$

The identity terms in $D\Psi = \log X + I$ cancel. The positive Hessian and the implicit function theorem make the minimizer smooth on $J \succ 0$. Boundary values require continuity of the minimum, rather than an assertion of interiority or uniqueness at singular J.

5 Concavity of the dual value

Take $J \succ 0$ and its unique positive minimizer W. Write $T = J + W$, $P = \Gamma J$, $A = \Omega_T$, $B = \Omega_W|_{\mathcal{B}}$, and $C = \Omega_P|_{\mathcal{B}}$. In this section the Hessians act on the real space of Hermitian matrices, with the Hilbert-Schmidt inner product.

The full objective $\Psi(\Gamma J) + f_J(W)$ has the block Hessian

$$\begin{pmatrix} VC^{-1}V^* - A^{-1} & -A^{-1}V \\ -V^*A^{-1} & B^{-1} - V^*A^{-1}V \end{pmatrix}. \tag{5.1}$$

The lower-right block $E = B^{-1} - V^*A^{-1}V$ is positive definite by the strict convexity proved in Section 4. Differentiating the stationary equation, or completing the quadratic form in the W-direction, gives the value Hessian

$$D^2G_\Lambda = VC^{-1}V^* - A^{-1} - A^{-1}VE^{-1}V^*A^{-1} \tag{5.2}$$

$$= VC^{-1}V^* - (A - VBV^*)^{-1}. \tag{5.3}$$

The last equality is the Woodbury inverse identity; it can also be checked by multiplying the two sides by $A - VBV^*$. All inverses in this identity act on the indicated Hilbert–Schmidt spaces.

Set $Z = A - VBV^*$. Since $T \succ W$ and the full Ω_W splits into diagonal and off-diagonal block spaces, $Z \succ 0$. The stationary equation and KM give

$$V^*ZV = V^*\Omega_T V - B \preceq C. \tag{5.4}$$

For any positive definite Z and any isometry V,

$$Z^{-1} \succeq V(V^*ZV)^{-1}V^*. \tag{5.5}$$

To prove this, $Z^{1/2}V(V^*ZV)^{-1/2}$ is an isometry, so its range projection is at most the identity. Congruence by $Z^{-1/2}$ gives the displayed inequality. Inversion of $V^*ZV \preceq C$ now implies

$$Z^{-1} \succeq V(V^*ZV)^{-1}V^* \succeq VC^{-1}V^*. \tag{5.6}$$

Consequently $D^2G_\Lambda \preceq 0$ on the positive definite cone. It follows by restriction to line segments that G_Λ is concave there. Approximate any semidefinite arguments by adding δI and use the continuity proved in Section 4. This extends concavity to the entire semidefinite cone.

Lemma 5.1 (Concavity of the dual value)**.** *For every fixed negative block-scalar Λ, the function G_Λ is continuous and concave on $J \succeq 0$.*

The qualifier "fixed" matters. The two outside masses $\operatorname{Tr} W_{00}$ and $\operatorname{Tr} W_{11}$ vary in the minimization. No concavity with those masses fixed is used or asserted.

6 Reduction to a scalar dual problem

Fix $x, z > 0$, and consider $\mathcal{C}_{x,z} = \{J \succeq 0 : \operatorname{Tr} J_{00} = x, \operatorname{Tr} J_{11} = z\}$. Every element has a finite convex decomposition into rank-one elements of the same set.

Mass-preserving rank reduction. If J has rank $q \geq 2$, Hermitian matrices on its support have real dimension $q^2 > 2$. Thus a nonzero supported Hermitian H has both block traces zero. It is traceless and hence indefinite. The line $J + tH$ has two finite semidefinite boundary endpoints on opposite sides of zero. Each endpoint has smaller rank; J is their convex combination. Recursing terminates after finitely many rank reductions.

A rank-one element has form $|u \oplus v\rangle\langle u \oplus v|$, where $\|u\|^2 = x$, $\|v\|^2 = z$. Independent block unitaries put it on two distinguished unit vectors, with active matrix

$$j = \begin{pmatrix} x & \sqrt{xz} \\ \sqrt{xz} & z \end{pmatrix}. \tag{6.1}$$

These unitaries leave G_Λ invariant. To show that unused directions do not reduce its value, average W over the four block-diagonal unitaries obtained by independently choosing a sign on each unused block subspace and acting as the identity on both distinguished vectors. Each of these unitaries preserves J. Convexity of f_J shows that the averaging cannot increase its value. The averaged W has active entries $u, v \geq 0$ and independent complementary positive blocks.

On each complementary block, $J = 0$; the two Ψ-terms cancel. Only $-\operatorname{Tr} \Lambda W \geq 0$ remains, and it is minimized by zero. Hence the rank-one value is exactly

$$g_{x,z}(\Lambda) = \min_{u,v \geq 0} \{f_{x,z}(u, v) - \lambda_0 u - \lambda_1 v\}, \tag{6.2}$$

$$f_{x,z}(u, v) = x \log x + z \log z + u \log u + v \log v - \Psi\left(j + \operatorname{diag}(u, v)\right). \tag{6.3}$$

Concavity and the mass-preserving decomposition give $G_\Lambda(J) \geq g_{x,z}(\Lambda)$ for every $J \in \mathcal{C}_{x,z}$.

Selecting the dual coefficients. Given $y, w > 0$, put $L = j + \operatorname{diag}(y, w)$, and choose

$$\lambda_0 = \log y - (\log L)_{00}, \qquad \lambda_1 = \log w - (\log L)_{11}. \tag{6.4}$$

Both are strictly negative. For example, with $v_* = (\sqrt{x}, \sqrt{z})^T$, $D = \operatorname{diag}(y, w)$, and $D_t = D + tI$, the resolvent formula gives

$$(D_t + v_* v_*^*)^{-1} = D_t^{-1} - \frac{D_t^{-1} v_* v_*^* D_t^{-1}}{1 + v_*^* D_t^{-1} v_*}. \tag{6.5}$$

The correction has strictly positive diagonal entries. Integrating the resolvent difference in the logarithm representation gives $(\log L)_{00} > \log y$ and $(\log L)_{11} > \log w$.

The function $f_{x,z}$ is convex by the preceding convexity proof for fixed $J = j$, including the boundary by continuity. Its gradient at (y, w) is (λ_0, λ_1). Therefore (y, w) globally minimizes the scalar dual objective. This identifies its value without assuming uniqueness at a singular central matrix.

7 The four-block entropy inequality

Theorem 7.1 (Four-block entropy inequality (TC4))**.** *For arbitrary compatible complex rectangular matrices,*

$$M = \begin{pmatrix} A & B & 0 \\ C & 0 & D \end{pmatrix}, \qquad (x, y, z, w) = (\|A\|_F^2, \|B\|_F^2, \|C\|_F^2, \|D\|_F^2). \tag{7.1}$$

assume $x + y + z + w = 1$, and put

$$L = \begin{pmatrix} x+y & \sqrt{xz} \\ \sqrt{xz} & z+w \end{pmatrix}. \tag{7.2}$$

Then

$$S(MM^*) \geq S(L) + xS(AA^*/x) + yS(BB^*/y) + zS(CC^*/z) + wS(DD^*/w). \tag{7.3}$$

Zero-weight terms are defined as zero.

Proof. Positive masses. Put $J = \begin{pmatrix} A \\ C \end{pmatrix} \begin{pmatrix} A^* & C^* \end{pmatrix}$ and $W = BB^* \oplus DD^*$. Thus $J+W = MM^*$, the central block traces are x, z, and the outer block traces are y, w. Choose Λ as in Section 6. By the definition of the minimum and the rank-one lower bound,

$$\begin{aligned} \Psi(\Gamma J) + \Psi(W) - \Psi(J+W) - \operatorname{Tr}\Lambda W &\geq G_\Lambda(J) \\ &\geq f_{x,z}(y,w) - \lambda_0 y - \lambda_1 w. \end{aligned} \tag{7.4}$$

The linear terms cancel because $\operatorname{Tr}\Lambda W = \lambda_0 y + \lambda_1 w$. We obtain

$$\begin{aligned} \Psi(AA^*) + \Psi(BB^*) + \Psi(CC^*) + \Psi(DD^*) - \Psi(MM^*) \\ \geq x\log x + y\log y + z\log z + w\log w - \Psi(L). \end{aligned} \tag{7.5}$$

For a positive semidefinite matrix R with trace $a > 0$,

$$\Psi(R) = a\log a - aS(R/a). \tag{7.6}$$

Substitution gives TC4, since MM^* and L have trace one. The matrices J and W may already be singular: Lemma 5.1 and convexity on the closed cone cover them.

Zero masses. Perturb any zero block to a nonzero rectangular block, and normalize all four blocks by their total Frobenius norm. Apply the positive-mass inequality and take the perturbation to zero. The weighted term is continuous at zero because

$$aS(R/a) = a\log a - \Psi(R) \longrightarrow 0 \qquad (R \to 0). \tag{7.7}$$

All remaining terms are continuous by continuity of $t\log t$ on $[0,\infty)$. A block with no columns can first be padded with zero columns, which changes neither its Gram matrix nor MM^*. Thus all degenerate configurations are included. □

Equality example. For any density matrix τ, take $A = \sqrt{x}\sqrt{\tau}$, $B = \sqrt{y}\sqrt{\tau}$, $C = \sqrt{z}\sqrt{\tau}$, and $D = \sqrt{w}\sqrt{\tau}$. Then $MM^* = L \otimes \tau$, and both sides of TC4 are $S(L) + S(\tau)$. Thus equality is attained, and the four internal coefficients cannot all be increased.

8 A normal form for every qubit channel

Let $\Phi : M_2 \to M_2$ be a channel. Its Hilbert-Schmidt adjoint Φ^* is completely positive and unital. For output basis matrix units, write

$$\begin{gathered} P = \Phi^*(E_{00}), \qquad Q = \Phi^*(E_{01}), \qquad R = \Phi^*(E_{11}), \\ \begin{pmatrix} P & Q \\ Q^* & R \end{pmatrix} \succeq 0, \qquad P + R = I. \end{gathered} \tag{8.1}$$

Choosing the output basis. Consider the homogeneous quartic

$$p(s,t) = \det\bigl(s^2\Phi^*(E_{10}) + st[\Phi^*(E_{11}) - \Phi^*(E_{00})] - t^2\Phi^*(E_{01})\bigr). \tag{8.2}$$

It either vanishes identically or has a zero in $\mathbb{CP}^1$, by the fundamental theorem of algebra (including the point at infinity). Choose such (s,t) with $|s|^2 + |t|^2 = 1$, and put

$$u = (\bar{s}, \bar{t})^T, \quad v = (-t, s)^T, \quad |v\rangle\langle u| = \begin{pmatrix} -st & -t^2 \\ s^2 & st \end{pmatrix}. \tag{8.3}$$

The vectors are orthonormal. In the reordered output basis (v, u), the off-diagonal dual image has determinant zero. Since its input matrix is two by two, its rank is at most one. Output conjugation preserves all entropies and Holevo quantities, so we work in this basis.

Rank-one block factorization. If $Q \neq 0$, write $Q = \sigma pq^*$, $\sigma > 0$, with p, q unit vectors. Write P^+ for the inverse of P on its support, extended by zero on its kernel. Block positivity ensures $p \in \operatorname{ran} P$, $q \in \operatorname{ran} R$, and $Q^*P^+Q \preceq R$. Let

$$\alpha = p^*P^+p > 0, \qquad a = p/\sqrt{\alpha}, \qquad c = \sigma\sqrt{\alpha}\, q. \tag{8.4}$$

Then $Q = ac^*$, $aa^* \preceq P$, and $cc^* \preceq R$. The first order inequality follows from Cauchy-Schwarz after conjugating by $P^{+1/2}$ on its support; the second is the generalized Schur complement just stated.

Take finite spectral factorizations of the residual matrices:

$$P = aa^* + \sum_j b_j b_j^*, \qquad R = cc^* + \sum_k d_k d_k^*. \tag{8.5}$$

When $Q = 0$, simply take $a = c = 0$ and factor P, R. In both cases define

$$K = \begin{pmatrix} a^* \\ c^* \end{pmatrix}, \quad K_{0j} = \begin{pmatrix} b_j^* \\ 0 \end{pmatrix}, \quad K_{1k} = \begin{pmatrix} 0 \\ d_k^* \end{pmatrix}. \tag{8.6}$$

Their dual action on the four matrix units equals that of Φ^*, hence they represent Φ. Moreover

$$aa^* + cc^* + \sum_j b_j b_j^* + \sum_k d_k d_k^* = I. \tag{8.7}$$

This verifies trace preservation and includes singular P, R, zero Q, and arbitrary Kraus rank. A quartic root is chosen separately for each channel, so no continuous choice is required.

9 The single-channel convex roof

Use the normal form from Section 8. For a qubit input state ρ, put

$$\begin{aligned} x = a^*\rho a, \qquad z = c^*\rho c, \qquad t = \operatorname{Tr} P\rho, \\ y = t - x, \qquad w = 1 - t - z. \end{aligned} \tag{9.1}$$

$$L(\rho) = \begin{pmatrix} t & \sqrt{xz} \\ \sqrt{xz} & 1-t \end{pmatrix}. \tag{9.2}$$

All four masses are nonnegative, by $P \succeq aa^*$, $R \succeq cc^*$. Moreover $L(\rho) \succeq 0$ because its determinant is $xw + yz + yw \geq 0$, and its trace is one.

Lemma 9.1 (A single-channel upper bound)**.** *For every qubit input state,* $\widehat{S}_\Phi(\rho) \leq S(L(\rho))$.

Proof. If ρ is mixed, it is positive definite. The real space of Hermitian two by two matrices has dimension four. The three homogeneous constraints

$$\operatorname{Tr} H = 0, \quad \operatorname{Tr}(aa^*H) = 0, \quad \operatorname{Tr}(cc^*H) = 0 \tag{9.3}$$

therefore admit a nonzero Hermitian solution. A nonzero traceless Hermitian two by two matrix is indefinite. The line $\rho + sH$ meets the boundary of the density-matrix ball in two states ρ_-, ρ_+, with $s_- < 0 < s_+$. Both are rank one and hence pure. Write ρ as their convex combination. In each component the values x, z are unchanged.

This also covers linearly dependent or zero effects: fewer independent constraints leave at least as many directions. If ρ is already pure, use its one-term decomposition. Thus in every case there is a pure ensemble $\rho = \sum_i p_i\rho_i$ preserving x, z individually.

For a pure component $\rho_i = |\psi_i\rangle\langle\psi_i|$, the actual channel output is

$$\Phi(\rho_i) = \begin{pmatrix} t_i & a^*\rho_i c \\ c^*\rho_i a & 1-t_i \end{pmatrix}, \qquad |a^*\rho_i c|^2 = xz. \tag{9.4}$$

The last equality is the product of the two squared amplitudes. Its output eigenvalues therefore equal those of

$$L_i = \begin{pmatrix} t_i & \sqrt{xz} \\ \sqrt{xz} & 1-t_i \end{pmatrix}. \tag{9.5}$$

Although the phases of the actual outputs can vary, entropy depends only on eigenvalues. The comparison matrices themselves satisfy $\sum_i p_i L_i = L(\rho)$, since $\sum_i p_i t_i = t$. Entropy concavity now gives

$$\widehat{S}_\Phi(\rho) \leq \sum_i p_i S(\Phi(\rho_i)) = \sum_i p_i S(L_i) \leq S(L(\rho)). \tag{9.6}$$

This proof does not preserve all four masses; only x, z are fixed across components. It does not claim that the selected two-state ensemble is optimal. In particular it does not conflict with channels requiring more than two signals to attain their optimized Holevo quantity. □

The two-dimensional input assumption is essential here: density-matrix boundary points on the chosen line are pure only in dimension two. The theorem concerns input and output dimension two for the first channel.

10 The arbitrary partner channel

Let $|\psi\rangle \in \mathbb{C}^2 \otimes \mathcal{Y}$ be a unit pure input, and set $\rho_A = \mathrm{Tr}_{\mathcal{Y}}\, \pi_\psi$, $\rho_B = \mathrm{Tr}_{\mathbb{C}^2}\, \pi_\psi$. For each normal-form vector $f \in \{a, c, b_j, d_k\}$, define

$$|\eta_f\rangle = (\langle f| \otimes I)|\psi\rangle. \tag{10.1}$$

Take a Stinespring isometry $U : \mathcal{Y} \to \mathcal{Z} \otimes \mathcal{E}$ for Ω. Let X_f be the coefficient matrix of $U\eta_f$ in fixed bases of $\mathcal{Z}$ and $\mathcal{E}$, with rows indexed by $\mathcal{Z}$. Then

$$X_f X_f^* = \Omega(|\eta_f\rangle\langle\eta_f|), \qquad \|X_f\|_F^2 = \|\eta_f\|^2. \tag{10.2}$$

Tracing out the Kraus environment of Φ and the Stinespring environment of Ω gives exactly

$$\begin{gathered}(\Phi \otimes \Omega)(\pi_\psi) = MM^*, \\ M = \begin{pmatrix} X_a & [X_{b_1}\ X_{b_2}\ \cdots] & 0 \\ X_c & 0 & [X_{d_1}\ X_{d_2}\ \cdots] \end{pmatrix}.\end{gathered} \tag{10.3}$$

The common central column block corresponds to the single Kraus branch K; each side block concatenates all branches confined to its row. The masses are precisely x, y, z, w defined at ρ_A in Section 9, so the scalar comparison is $L(\rho_A)$.

By TC4,

$$\begin{aligned}S((\Phi \otimes \Omega)(\pi_\psi)) \geq & S(L(\rho_A)) + xS(X_a X_a^*/x) + zS(X_c X_c^*/z) \\ & + yS(BB^*/y) + wS(DD^*/w),\end{aligned} \tag{10.4}$$

where $B = [X_{b_j}]$ and $D = [X_{d_k}]$. Entropy concavity separates the concatenated blocks: for example, if $y > 0$,

$$yS(BB^*/y) \geq \sum_j \|\eta_{b_j}\|^2 S\left(\Omega\left(\frac{|\eta_{b_j}\rangle\langle\eta_{b_j}|}{\|\eta_{b_j}\|^2}\right)\right). \tag{10.5}$$

Omit zero-weight branches. The Kraus completeness equation proves the exact marginal identity

$$\sum_f |\eta_f\rangle\langle\eta_f| = \rho_B. \tag{10.6}$$

Indeed, expand $|\psi\rangle = \sum_{i=0}^{1} |i\rangle \otimes |\psi_i\rangle$, and use $\sum_f f f^* = I$ in the resulting coefficients. The normalized nonzero η_f consequently form a pure-state ensemble of ρ_B, rather than of a surrogate state.

The total internal entropy term is at least $\widehat{S}_\Omega(\rho_B)$. Applying Lemma 9.1 gives

$$\begin{aligned}S((\Phi \otimes \Omega)(\pi_\psi)) &\geq S(L(\rho_A)) + \widehat{S}_\Omega(\rho_B) \\ &\geq \widehat{S}_\Phi(\rho_A) + \widehat{S}_\Omega(\rho_B).\end{aligned} \tag{10.7}$$

This includes zero masses via the TC4 boundary convention. No assumption about the partner's additivity, Kraus rank, or dimensions has entered.

11 Completion of the proof

The convex roof $\widehat{S}_\Phi$ is convex. For a direct proof, select pure decompositions within ε of the infimum for each state, concatenate them with the mixing probabilities, and let $\varepsilon \downarrow 0$.

Take any pure decomposition $\omega = \sum_i p_i \pi_{\psi_i}$ of a bipartite mixed input. The pure-input inequality from Section 10 and convexity give

$$\begin{aligned} \sum_i p_i S((\Phi \otimes \Omega)(\pi_{\psi_i})) &\geq \sum_i p_i \widehat{S}_\Phi((\pi_{\psi_i})_A) + \sum_i p_i \widehat{S}_\Omega((\pi_{\psi_i})_B) \\ &\geq \widehat{S}_\Phi(\omega_A) + \widehat{S}_\Omega(\omega_B). \end{aligned} \tag{11.1}$$

Taking the infimum over all such decompositions proves (1.4).

Because both channels are trace-preserving, the output marginals of $(\Phi \otimes \Omega)(\omega)$ are $\Phi(\omega_A)$ and $\Omega(\omega_B)$. Entropy subadditivity therefore gives

$$\begin{aligned} S((\Phi \otimes \Omega)(\omega)) - \widehat{S}_{\Phi\otimes\Omega}(\omega) &\leq S(\Phi(\omega_A)) - \widehat{S}_\Phi(\omega_A) \\ &\quad + S(\Omega(\omega_B)) - \widehat{S}_\Omega(\omega_B) \\ &\leq \chi(\Phi) + \chi(\Omega). \end{aligned} \tag{11.2}$$

Supremizing over ω proves the upper bound. For the lower bound, take any finite signal ensemble for each channel and form the independent product ensemble. Entropy additivity on product states makes its Holevo information the sum of the two original ensemble values. Taking ensembles approaching both suprema yields the opposite inequality.

This proves Theorem 1.1.

Corollary 11.1 (Tensor powers)**.** *For every qubit-to-qubit channel Φ and positive integer k,*

$$\chi(\Phi^{\otimes k}) = k\chi(\Phi). \tag{11.3}$$

Proof. Apply Theorem 1.1 with partner $\Phi^{\otimes(k-1)}$ and induct on k. □

The standard Holevo–Schumacher–Westmoreland capacity formula then gives $C(\Phi) = \chi(\Phi)$. This operational consequence uses the coding theorem; that theorem is not required for (1.5).

12 Conclusion

The proof reduces strong Holevo additivity to the four-block entropy inequality of Theorem 7.1. A stationary Gram identity yields a differential inequality for a positive matrix curve. Comparison with the endpoint geometric mean, followed by integration, gives the operator estimate used to establish concavity of the dual value function. A rank-one decomposition preserving the central masses reduces this value to a scalar problem. Choosing its multipliers at the prescribed outer masses proves TC4, including singular blocks and zero weights.

For every qubit-to-qubit channel, an output-basis choice and a positive block factorization give the required Kraus form with one common branch. A pure-state decomposition preserving two expectations supplies the single-channel convex-roof bound. For a bipartite pure input, the same Kraus form realizes the joint output as a four-block coefficient matrix, while Kraus completeness reconstructs the actual partner marginal. TC4 therefore gives the pure-input entropy bound for an arbitrary finite-dimensional partner.

Convexity extends this bound to mixed inputs and yields

$$\widehat{S}_{\Phi\otimes\Omega}(\omega) \geq \widehat{S}_{\Phi}(\omega_A) + \widehat{S}_{\Omega}(\omega_B). \tag{12.1}$$

Entropy subadditivity gives the corresponding upper bound for optimized Holevo information, and product ensembles give the reverse inequality. Thus, for every channel $\Phi : M_2(\mathbb{C}) \to M_2(\mathbb{C})$ and every channel $\Omega : M_n(\mathbb{C}) \to M_m(\mathbb{C})$, with arbitrary positive integers n, m,

$$\boxed{\chi(\Phi \otimes \Omega) = \chi(\Phi) + \chi(\Omega).} \tag{12.2}$$

In particular, $\chi(\Phi^{\otimes k}) = k\chi(\Phi)$ for every positive integer k. The first channel is arbitrary within input and output dimension two, and the partner is arbitrary in finite dimensions.